University of Science and Technologies Hanoi
Information Communication and Technologies department

Thesis for the Degree of Master of Science in Data Science & Robotics

MAVEN-01: A TELEOPEATED ROVER PLATFORM FOR PLANETARY EXPLORATION

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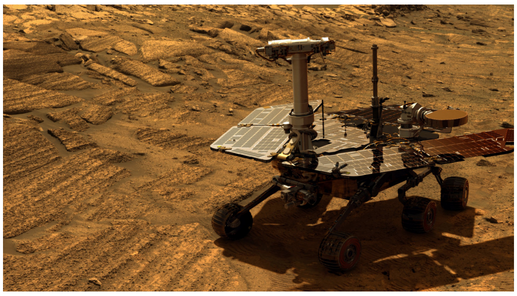
Abstract

This abstract summarizes the design, development, and functional improvements of MAVEN-01, a rover platform whose primary objective is to provide a highly customizable mobile system for planetary exploration missions on Earth in the present and on other planets in the future. The report elaborates on the rover's design and the techniques used in its development, such as 3D printing. Furthermore, in addition to the mechanical frame design, this paper also focuses on the electrical system implemented in the rover. Due to limited budget and time constraints, we selected the popular ESP32 microcontroller, which features built-in Wi-Fi and Bluetooth capabilities for teleoperation purposes. Other components were also carefully chosen to balance performance and cost. For example, the DHT11 or DHT12 sensors are suitable for temperature and humidity measurement, while the HC-SR04 sensor is used for distance measurement. The paper also discusses practical challenges, including durability, reliability, and ease of maintenance.

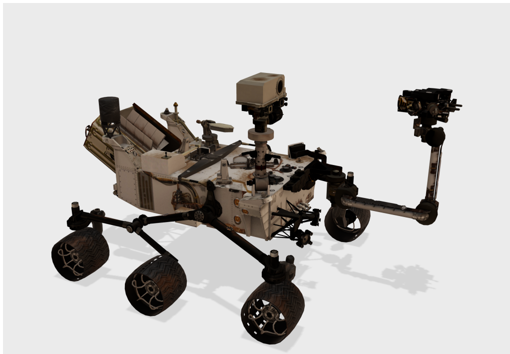
1. Introduction

Alongside exploration tasks, the importance of developing teleoperated mobile systems has been recognized for a long time. Historically, humans have conducted exploration missions themselves, which remains effective for many low-risk and specific tasks. However, for higher-risk and more hazardous missions—such as volcanic surveys or planetary exploration—researchers rely on specialized vehicles known as *rovers*. For example, NASA launched the Opportunity in 2003 and the Curiosity in 2012, two of the longest-operating missions on Mars, aimed at determining whether the planet could have supported life.

Although each rover was designed for different specific missions and utilized distinct technologies, they share several common structural features. Both systems were designed to be controlled and monitored from NASA headquarters on Earth. Additionally, they were engineered for high reliability and strong terrain adaptability—key factors that serve as motivation for the development of our own rover platform.



(a) Opportunity Rover



(b) Curiosity Rover

Figure 1: Curiosity and Opportunity Rovers hold record for two longest operating time, 14 years (5100 Sols) for Opportunity and 13.7 years (4863 Sols) for Curiosity.

This section introduces our rover platform, MAVEN-01, which represents the first iteration of our design. We are currently continuing its development to improve performance, durability, and reliability. Through multiple prototypes and design iterations—including unsuccessful attempts—MAVEN-01 has evolved with a clear objective: to provide a low-cost rover platform for research and experimental purposes using 3D printing technology. The rover is designed to support customization for specific tasks, featuring multiple mounting holes on the top surface to accommodate additional components and modules.

Maven-01 a was design for research, testing or educational purpose so every components are price affordable for students and laboratory. The durability and easy maintenance will be also focused on for budget saving and reliability.

Place holder for MAVEN 01 image

1.1 System Architecture

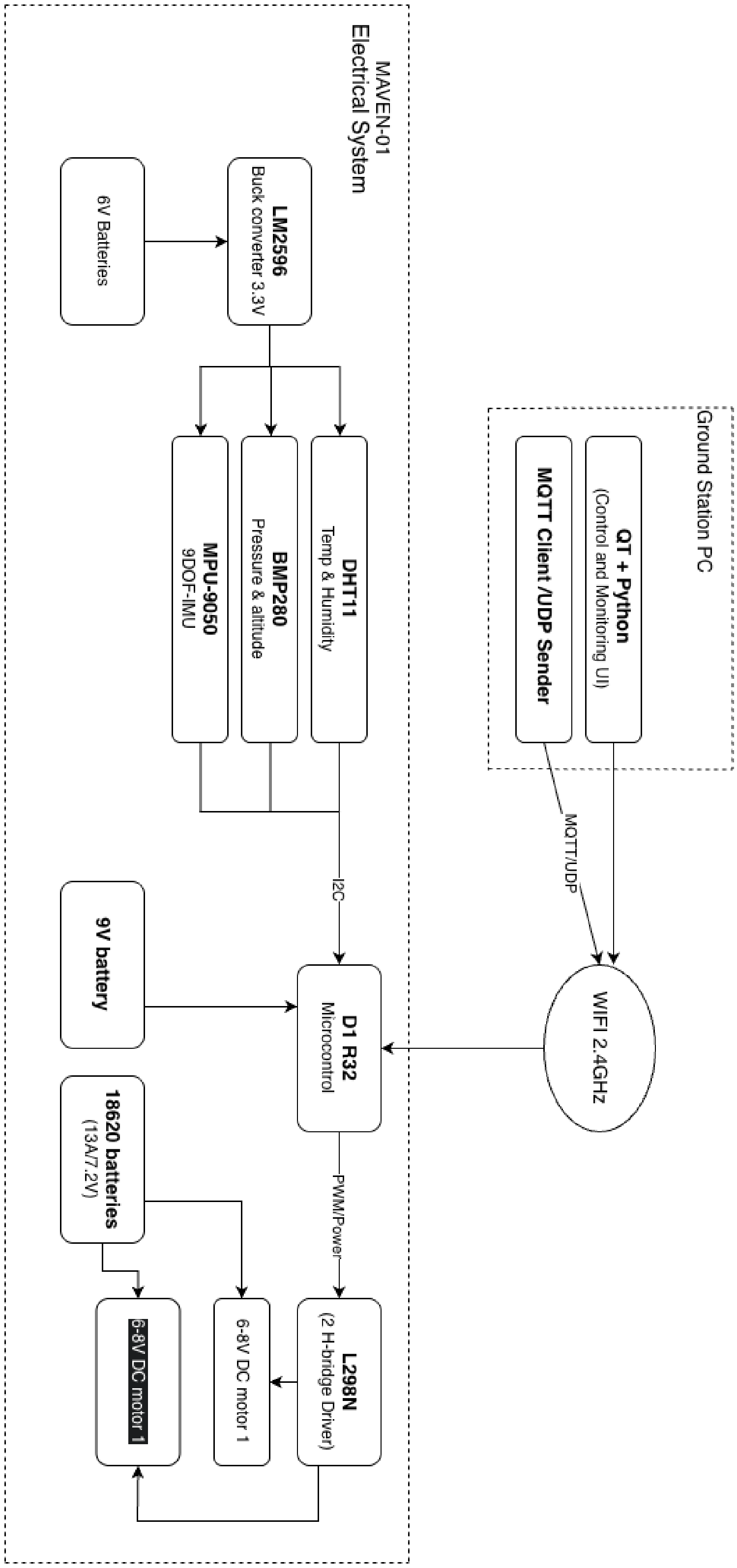


Figure 2: Maven 01 UML

2. Design and Development

2.1 Structure Design Overview

2.1.1 First version

MAVEN-01 was first designed as a two-tracked, tank-type vehicle platform using two 6V TT gear motors for each track. The size of the rover frame platform is 16.8 × 11 cm. The reason why the first body was small is due to the space limitation of the printing plate area of the Bambu Lab A1 Mini, which is our 3D printer. During the process, there were two material options: PLA and PETG. PETG is cheaper than PLA and offers higher durability and better heat resistance, but it is harder to print and requires careful storage due to its humidity-sensitive properties.

We chose PLA for its easier printing process and better print consistency, which is more suitable for rapid prototyping and iterative design. PETG might be chosen for more specific missions that require higher durability in harsher environments.

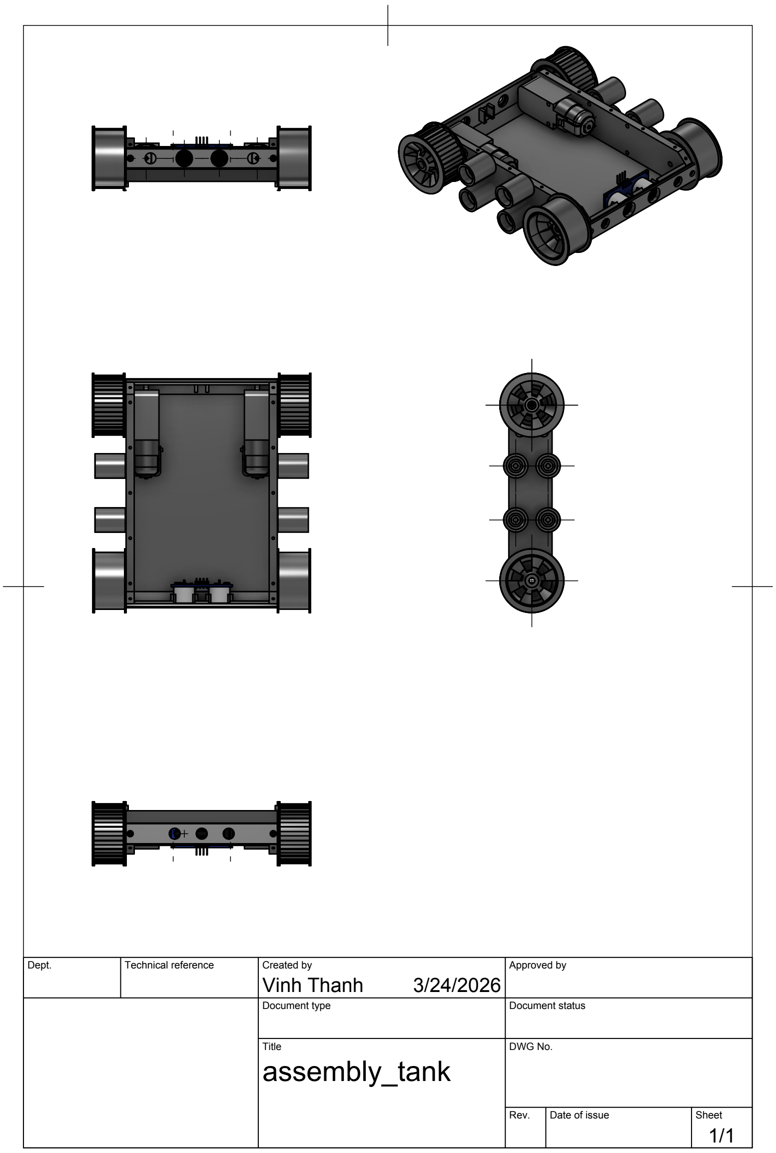


Figure 3: Blueprint of MAVEN-01 first prototype

Disadvantages However, there are some disadvantages in the first design that should be carefully considered. Due to the low ground clearance, this design cannot operate effectively in rugged terrain.

Another disadvantage is that the track can slip off the wheels due to the unstable drive wheel. In the first design, M4×30 screws and nuts are used to mount the drive wheel and front wheel. This design has a weakness: during testing, the screws wore into the wheel holes on the body, which caused the drive wheels to loosen over time, making the track structure weaker.

The last weakness of this design is two engines are mounted inside the rover body, which occupy a large space, we can put more components inside base body. During operation time, two DC motor can generate heat which can effect on micro-control board, also not ideal for PLA body due to not limited heat resistance.

Feature	**PLA (Polylactic Acid)**	**PETG (Polyethylene Terephthalate Glycol)**
Printability	Very Easy. Low warping, no enclosure needed.	Moderate. Requires higher temps; prone to stringing.
Strength	High tensile strength but brittle.	High impact resistance and more flexible.
Temp. Resistance	Low (softens at 70°C). Not for hot environments.	Moderate (softens at 85°C). Better for motors/sunlight.
Durability	Low. Can degrade outdoors over long periods.	High. UV resistant and chemically stable.
Post-Processing	Easy to sand and paint.	Difficult to sand due to its "gummy" nature.

Table 1: Comparison of PLA and PETG for Robotics Prototyping

2.1.2 Second Version

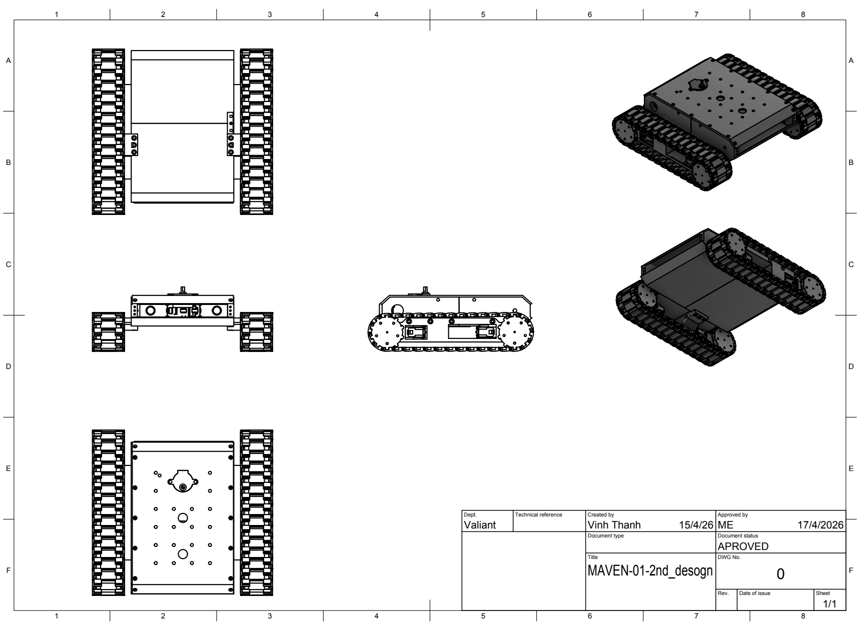


Figure 4: MAVEN-01 second prototype design using separate chassis with chained tracks (Single motor chassis)

In this version, we improve the design completely to fix the disadvantages of the first design. We move the track system into a separate module, which allows more space in the base rover body. With the separate track module, the ground clearance can be higher, allowing MAVEN-01 to adapt to different terrain types and rough surfaces. The L298N motor driver module and gear motor heating problems are also solved by the open-surface chassis design, which allows open-air ventilation to work as a cooling system.

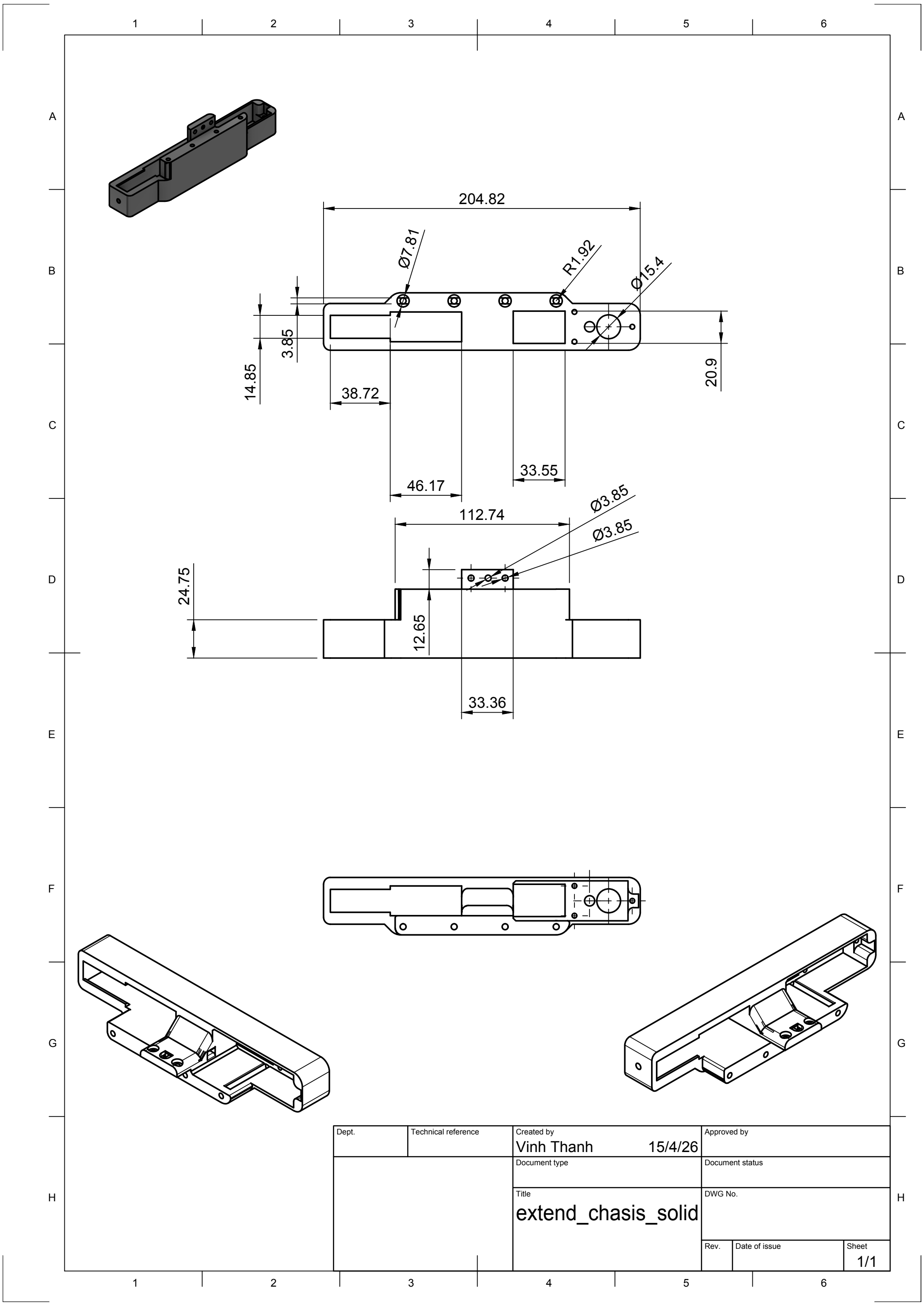


Figure 5: Single motor chassis blueprint for 6V TT gear motor with open surface for motor air cooling

Drive wheel design

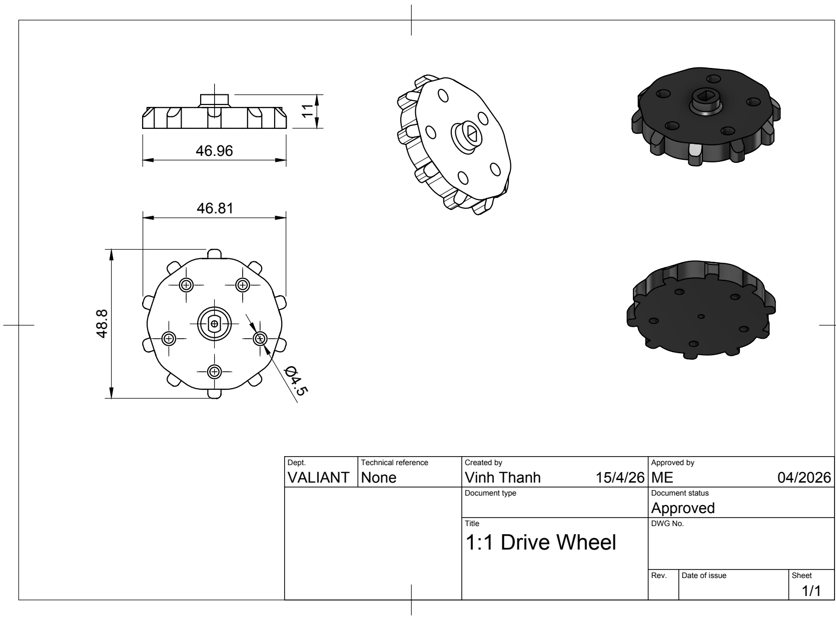


Figure 6: Driver wheels blueprint for MAVEN-01

Each TT motor will have two drive wheels act as anchors for track chains attached. Drive wheels can be attached info motor axises using glue or M2x10 to M2x12 tap screws.

Passive wheel design

Passive wheel were design to attach 6700zz bearing for smooth rotation, 6700zz has size of 15x10x4MM, mounted and screwed with M5x10 tap screw and M5 nuts as holding reinforcement.

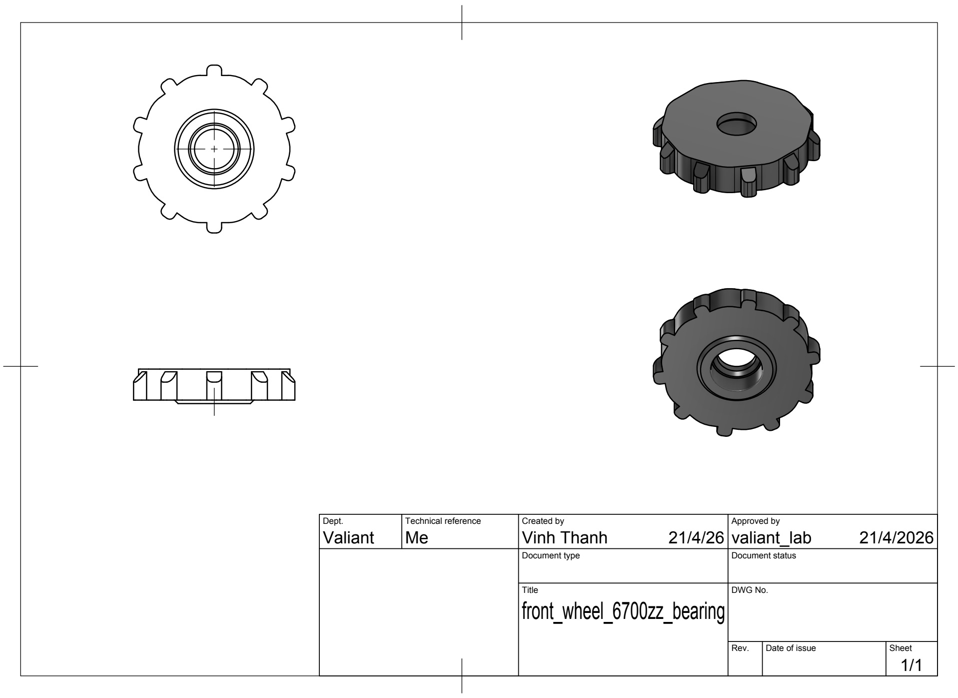


Figure 7: Front wheels with 6700zz bearingblueprint for MAVEN-01

Front axis

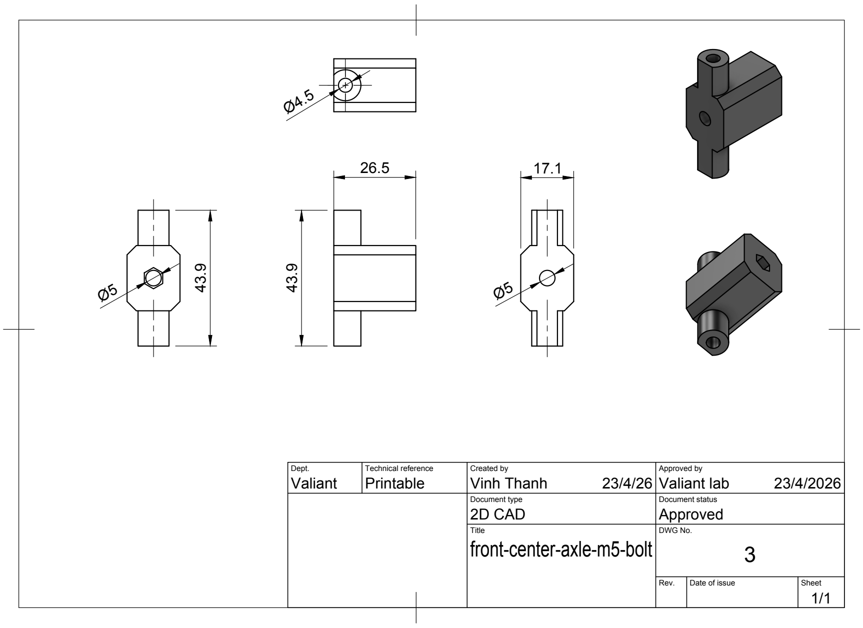


Figure 8: Front axis for front wheels for M5 bolts

Front wheels of Maven-01 are designed not to be attached directly to the chassis; they will be attached to a front wheel axis. The reason for separating this part from the chassis is that the front wheels are passive, which over time can be abraded. With the front wheels attached to the front axis instead of the chassis, they can be easily changed when worn out,and chassis replacement will not be required.

Track chain

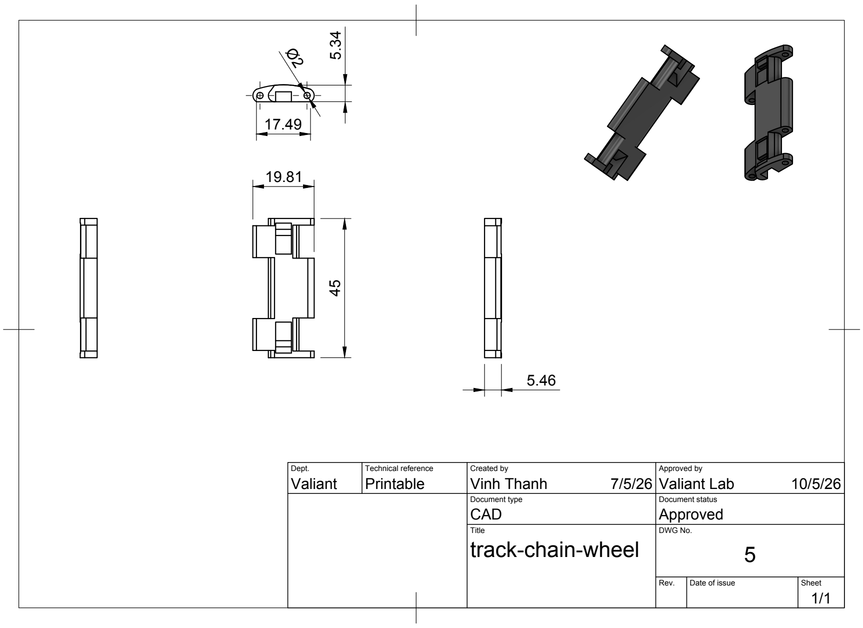


Figure 9: Front axis for front wheels for M5 bolts

MAVEN-01 uses an assembled chain design. In the first design, it used a rubber chain, but the wobbling axles caused the rubber track to become unstable, which could lead to derailment because there were no stable anchor points for the chain to mount onto both the front wheels and the drive wheels, which solved by this second design.

3. Electrical System

MAVEN-01 is designed to be an affordable autonomous vehicle platform for research and educational purposes, so as we chose low range price electrical components but still make sure the rover has solid performance by price.

Power source We use two 6C 18650 lithium batteries as the main power source for most of the system, include motor, sensors and lights. Lithium battery always has better power capacity and can be recharged. Two 6C 18650 batteries provide total 7.4V and 13A, which more than enough for the whole system requirement.

Motor Electrical Diagram

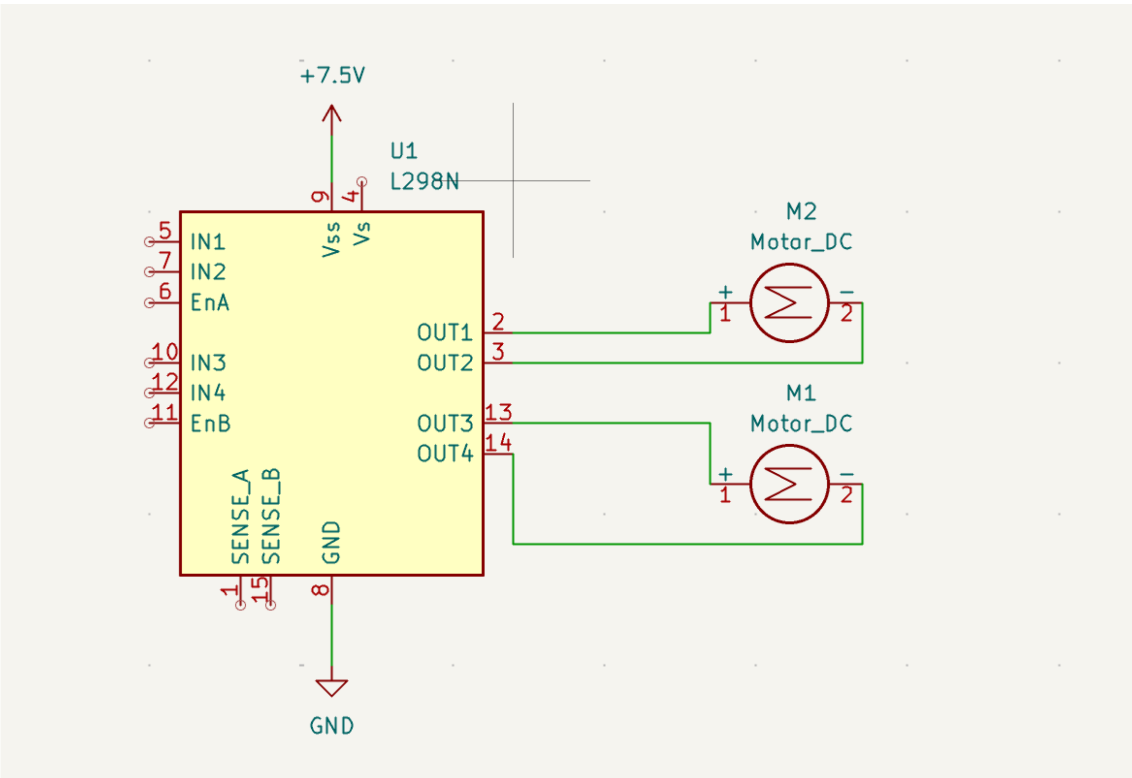


Figure 10: Diagram of connections between L289N with 2 6V TT gear motors

Two TT motor consume of total 6V due to the parallel wiring method through L298N, this motor driver allow power input in range from 5V to 12V, the maximum current <=2A, which are enough for two gear motors with total maximum current consume are 600mA. All calculation follow the Ohm's law, parallel connected module share constant voltage with cumulative currents.

Ohm's law

$$V = IR$$

Total current of parallel gear arrangement

$$I_{\text{total}} = I_1 + I_2 + \cdots + I_n$$

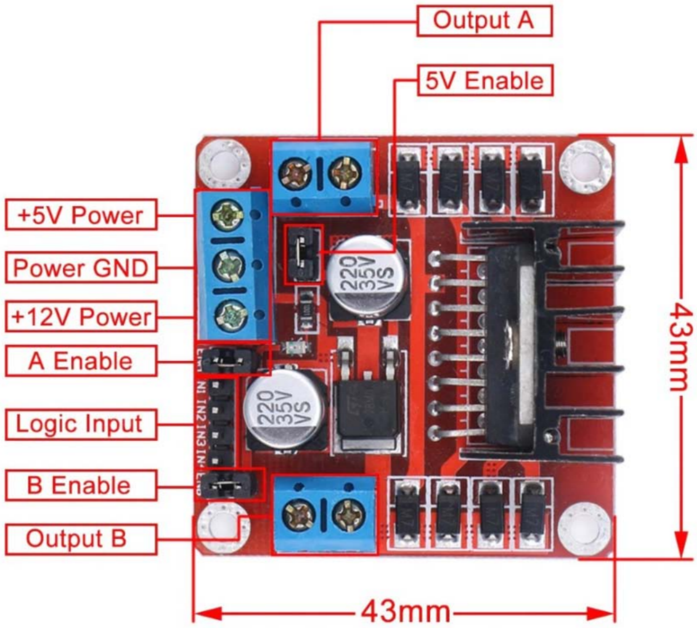


Figure 11: L298N H-bridge DC motor driver

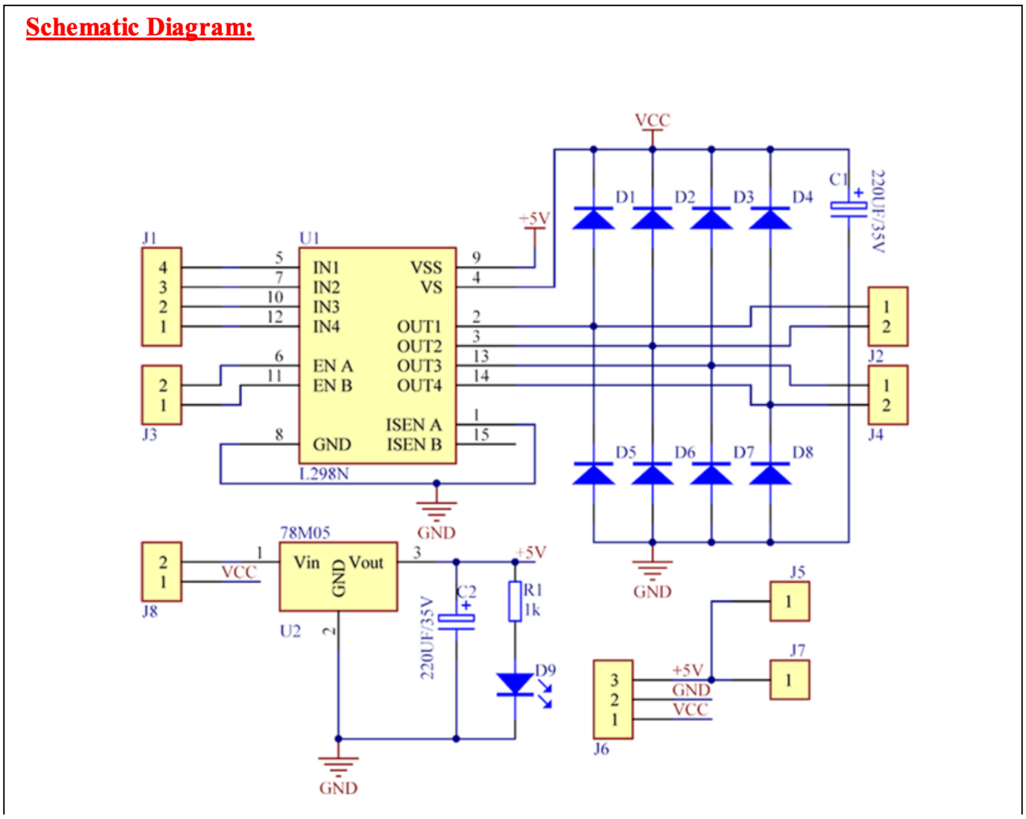


Figure 12: Schematic Diagram of L298N H-bridge DC motor driver

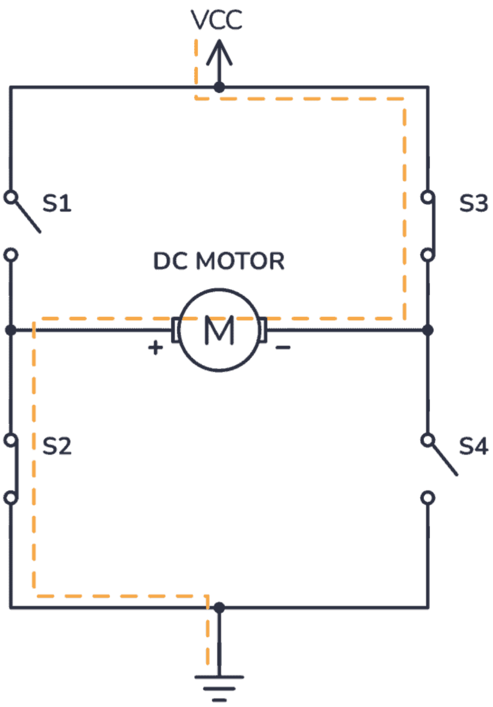


Figure 13: Schematic Diagram of L298N H-bridge DC motor driver

L298N has a structure called an H-bridge, which allows it to have a total of 4 cases of current direction to control DC motors. In Figure ??, we can see an example of a single motor in an H-bridge circuit. When `S3` and `S1` are closed, it allows DC current to run through the motor, making it spin counterclockwise. On the other hand, if `S1` and `S4` are closed while `S3` and `S2` are open, DC current will run through and make the motor spin in the clockwise direction. L298N has two H-bridge circuits, which in theory can control more than 2 DC motors, as long as the current limit is not exceeded.

During the development process, we tested with 1 and 2 DC motors on each bridge and concluded that the L298N can handle 4 DC motors with a total current of approximately 1.2A. However, the movement capability will be similar to using 2 motors since the L298N only has 2 H-bridges. For better control and manipulation, we decided to switch to a 4-channel motor driver module.

3.1 Control Protocol

MAVEN-01 was designed to be teleoperated, so it needs a control board that has some kind of wireless send-and-receive mechanism. Most of the time, radio signal is the first choice because of its wide range, stable connection, and relatively low power consumption. However, complete radio control requires a full set including a remote control, transmitter, receiver, etc., which increases the overall cost significantly.

In our project, we choose Wi-Fi as the main method to control the rover, not only for control but also to receive data from sensors attached to the rover. Because MAVEN-01 is designed for research purposes, Wi-Fi is a better choice. We only need a Wi-Fi built-in microcontroller board such as ESP32 or ESP8266. Sensor data are usually complex, so they require a high-bandwidth and low-latency signal to transmit. In this case, Wi-Fi performs better than radio, but the trade-off is higher latency and power consumption due to the higher frequency compare to normal radio signal.

3.2 Control Unit: WEMOS D1 R32 (ESP32 based)

ESP32 is a microcontroller board with built-in WiFi and Bluetooth that we chose as the main control unit. With its integrated WiFi module, it enables wireless control and sensor data transmission. Compared to other boards such as Arduino UNO R4 or ESP8266, ESP32 offers faster processing speed and more GPIO pins, allowing more sensors to be connected. It is also affordable.

Board Comparison

Feature	ESP32	Pico W	ESP8266	UNO R4 WiFi
CPU	Dual-core 240 MHz	Dual-core 133 MHz	Single-core 80 MHz	48 MHz Cortex-M4
RAM	520 KB	264 KB	160 KB	32 KB
Flash	4 MB	2 MB	4 MB	256 KB
WiFi	Yes	Yes	Yes	Yes
Bluetooth	Yes	No	No	No
GPIO	∼30	26	∼17	14
RTOS	Yes (FreeRTOS)	No	No	No
Real-time	Medium	High	Medium	High
Power	Medium	Low	Low	Low
Price (VND)	∼120k	∼180k	∼70k	∼700k–1M

Table 2: Comparison of ESP32, Pico W, ESP8266, and Arduino UNO R4 WiFi

ESP32 has many variants suitable for specific tasks. It also supports camera modules such as OV2640 or OV5640 for video streaming, making it suitable as an FPV camera for the rover. In MAVEN-01, we used an Arduino-based ESP32 board called WEMOS D1 R32. This board uses the same dual-core processor and features as the standard ESP32 from Espressif, but is easier to use due to its Arduino-like form factor.

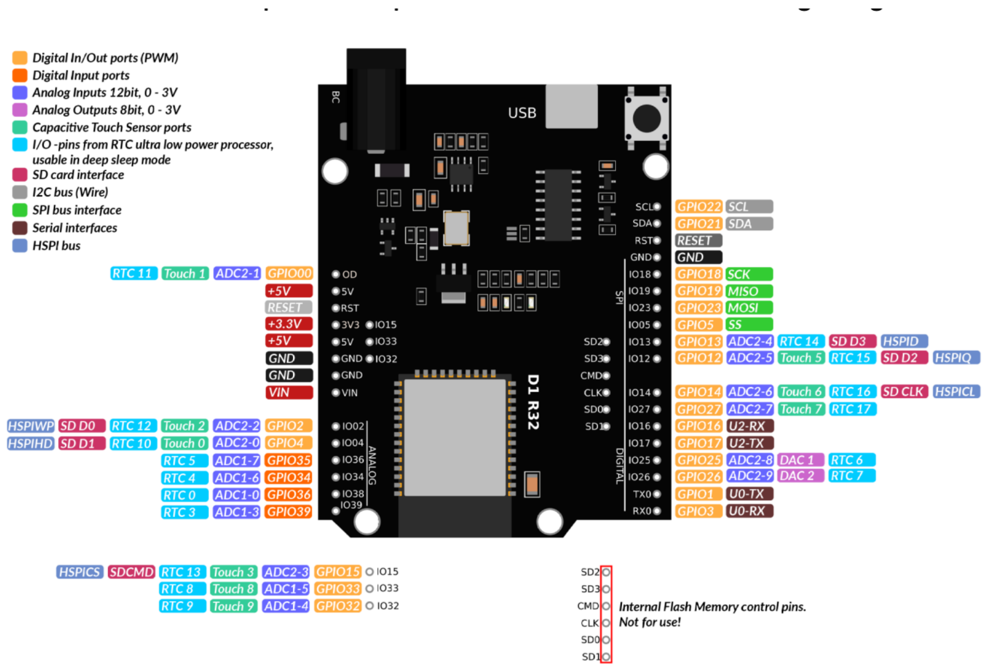


Figure 14: Wemos D1 R32 pins maps, more information about detailed pinout description in here

3.3 Temperature ana humidity sensor: DHT11

Overview

DHT11 Temperature & Humidity Sensor features a temperature & humidity sensor complex with a calibrated digital signal output. By using the exclusive digital-signal-acquisition technique and temperature & humidity sensing technology, it ensures high reliability and excellent long-term stability. This sensor includes a resistive-type humidity measurement component and an NTC temperature measurement component, and connects to a highperformance 8-bit microcontroller, offering excellent quality, fast response, anti-interference ability and cost-effectiveness.

Each DHT11 element is strictly calibrated in the laboratory that is extremely accurate on humidity calibration. The calibration coefficients are stored as programmes in the OTP memory, which are used by the sensor's internal signal detecting process. The single-wire serial interface makes system integration quick and easy. Its small size, low power consumption and up-to-20 meter signal transmission making it the best choice for various applications, including those most demanding ones. The component is 4-pin single row pin package. It is convenient to connect and special packages can be provided according to users' request.

Parameters	Conditions	Minimum	Typical	Maximum
Humidity				
Resolution		1%RH	1%RH	1%RH
			8 Bit	
Repeatability			±1%RH	
Accuracy	25℃		±4%RH	
	0-50℃			±5%RH
Interchangeability	Fully Interchangeable			
Measurement Range	0℃	30%RH		90%RH
	25℃	20%RH		90%RH
	50℃	20%RH		80%RH
Response Time (Seconds)	1/e(63%)25℃, 1m/s Air	6 S	10 S	15 S
Hysteresis			±1%RH	
Long-Term Stability	Typical		±1%RH/year	
Temperature				
Resolution		1℃	1℃	1℃
		8 Bit	8 Bit	8 Bit
Repeatability			±1℃	
Accuracy		±1℃		±2℃
Measurement Range		0℃		50℃
Response Time (Seconds)	1/e(63%)	6 S		30 S

Figure 15: DHT11 detailed specifications table, more information in here

Power and pins

DHT11's power supply is 3-5.5V DC. When power is supplied to the sensor, do not send any instruction to the sensor in within one second in order to pass the unstable status. One capacitor valued 100nF can be added between VDD and GND for power filtering.

Communication Process: Serial Interface (Single-Wire Two-Way)

Single-bus data format is used for communication and synchronization between MCU and DHT11 sensor. One communication process is about 4ms. Data consists of decimal and integral parts. A complete data transmission is 40bit, and the sensor sends higher data bit first. Data format: 8bit integral RH data + 8bit decimal RH data + 8bit integral T data + 8bit decimal T data + 8bit check sum. If the data transmission is right, the check-sum should be the last 8bit of "8bit integral RH data + 8bit decimal RH data + 8bit integral T data + 8bit decimal T data".

Overall Communication Process

When MCU sends a start signal, DHT11 changes from the low-power-consumption mode to the running-mode, waiting for MCU completing the start signal. Once it is completed, DHT11 sends a response signal of 40-bit data that include the relative humidity and temperature information to MCU. Users can choose to collect (read) some data. Without the start signal from MCU, DHT11 will not give the response signal to MCU. Once data is collected, DHT11 will change to the lowpower-consumption mode until it receives a start signal from MCU again.

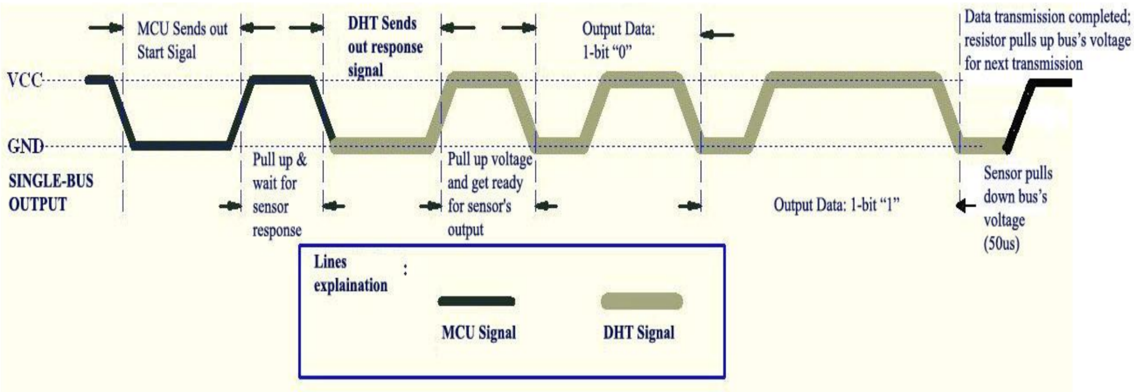


Figure 16: Detailed DHT 11 communication signal figure with MCU

Data Single-bus free status is at high voltage level. When the communication between MCU and DHT11 begins, the programme of MCU will set Data Single-bus voltage level from high to low and this process must take at least 18ms to ensure DHT's detection of MCU's signal, then MCU will pull up voltage and wait 20-40µs for DHT's response.

MCU send 'wake up' signal to DHT11

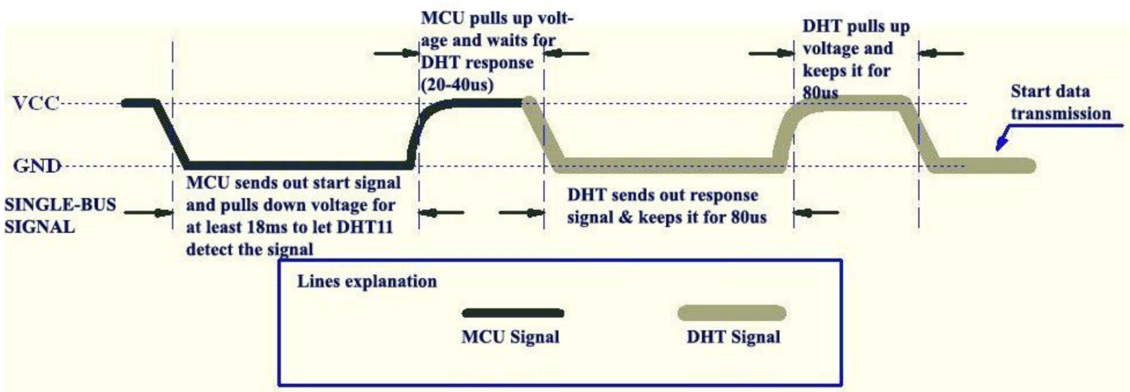


Figure 17: MCU Sends out Start Signal & DHT Responses

DHT11 'waves back' to the MCU

Once DHT detects the start signal, it will send out a low-voltage-level response signal, which lasts 80µs. Then the programme of DHT sets Data Single-bus voltage level from low to high and keeps it for 80µs for DHT's preparation for sending data. When DATA Single-Bus is at the low voltage level, this means that DHT is sending the response signal. Once DHT sent out the response signal, it pulls up voltage and keeps it for 80µs and prepares for data transmission. When DHT is sending data to MCU, every bit of data begins with the 50µs low-voltage-level and the length of the following high-voltage-level signal determines whether data bit is "0" or "1" ,see Figures 18 and 19.

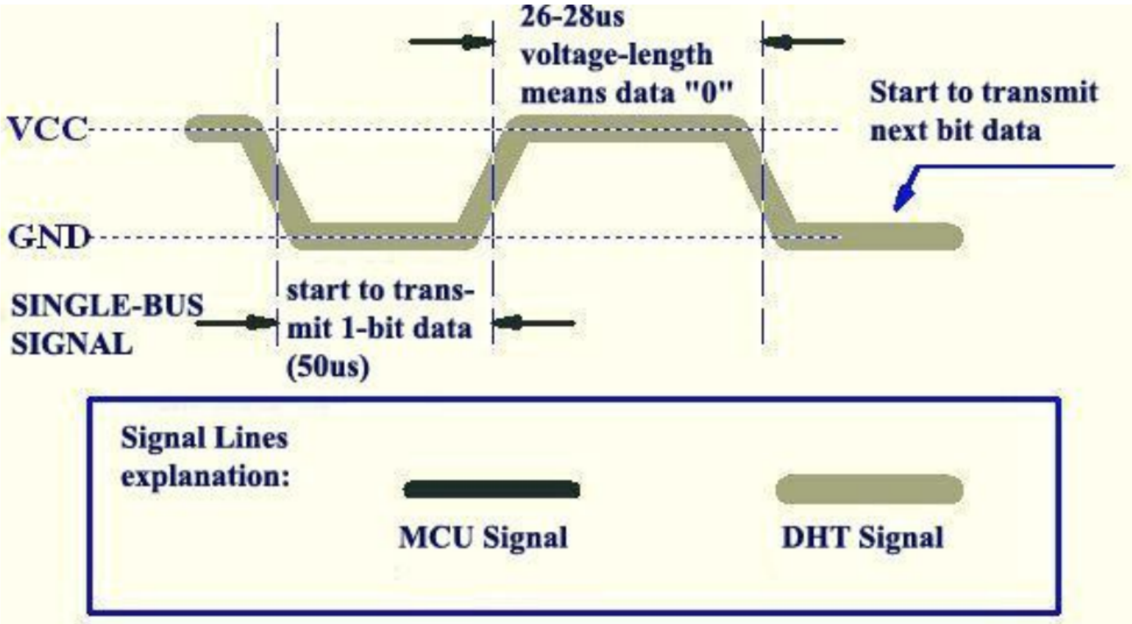


Figure 18: Data "0" Indication

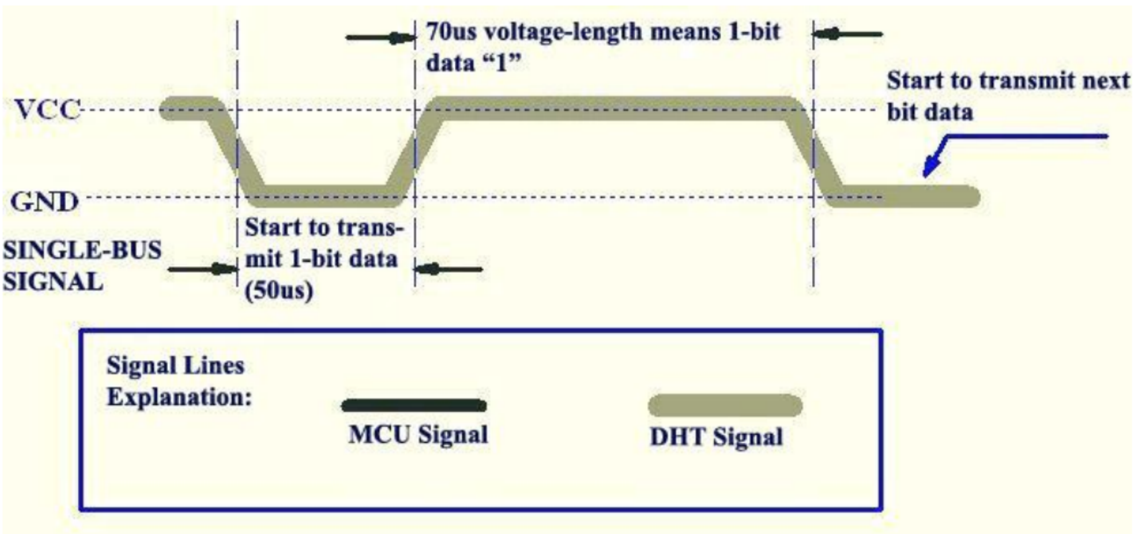


Figure 19: Data "1" Indication

Electrical Characteristics

VDD=5V, T = 25℃ (unless otherwise stated)

	Conditions	Minimum	Typical	Maximum
Power Supply	DC	3V	5V	5.5V
Current Supply	Measuring	0.5mA		2.5mA
	Average	0.2mA		1mA
	Standby	100uA		150uA
Sampling period	Second	1		

Figure 20: DHT11's electrical characteristics table

3.4 Distance measurement sensors: HC-SR04

Ultrasonic is an excellent way of figuring out what's in the immediate vicinity of your Arduino. The basics of using ultrasound are like this: you shoot out a sound, wait to hear it echo back, and if you have your timing right, you'll know if anything is out there and how far away it is. This is called echolocation and it's how bats and dolphins find objects in the dark and underwater, though they use lower frequencies than you can use with your Arduino.Figure-1 show the working principal of ultrasonic ranging concept.

HC-SR04 Ultrasonic Sensor is a very affordable proximity/distance sensor that has been used mainly for object avoidance in various robotics projects. It has also been used in turret applications, water level sensing, and even as a parking sensor. This module is the second generation of the popular HC-SR04 Low Cost Ultrasonic Sensor. Unlike the first generation HC-SR04 that can only operate between 4.8V 5V DC, the improved version has wider input voltage range, allow it to work with controller operates on 3.3V.HC-SR04 ultrasonic sensor provides a very low-cost and easy method of distance measurement. It measures distance using sonar, an ultrasonic (well above human hearing) pulse (40KHz) is transmitted from the unit and distance-to-target is determined by measuring the time required for the echo return. This sensor offers excellent range accuracy and stable readings in an easy-to-use package. An on board 2.54mm pitch pin header allows the sensor to be plugged into a solderless breadboard for easy prototyping

Module specification

Electrical Parameters	Value
Operating Voltage	3.3Vdc ~ 5Vdc
Quiescent Current	<2mA
Operating Current	15mA
Operating Frequency	40KHz
Operating Range & Accuracy	2cm ~ 400cm (1in ~ 13ft) ± 3mm
Sensitivity	-65dB min
Sound Pressure	112dB
Effective Angle	15°
Connector	4-pins header with 2.54mm pitch
Dimension	45mm x 20mm x 15mm
Weight	9g

Figure 21: HC-SR04 specifications table

The timing diagram, Figure 22 is shown below. You only need to supply a short 10uS pulse to "Trigger Input" pin to start the ranging. The module will send out 8-cycles burst of ultrasound at 40KHz and raise its "Echo" 6 pin, refer to Figure 23. The echo is a distance object that is pulse width and the range in proportion. You can calculate the range through the time interval between sending trigger signal and receiving echo signal. Formula: µS / 58 = centimeters or µS / 148 =inch or: the range = high level time * sound velocity (340m/s) / 2;

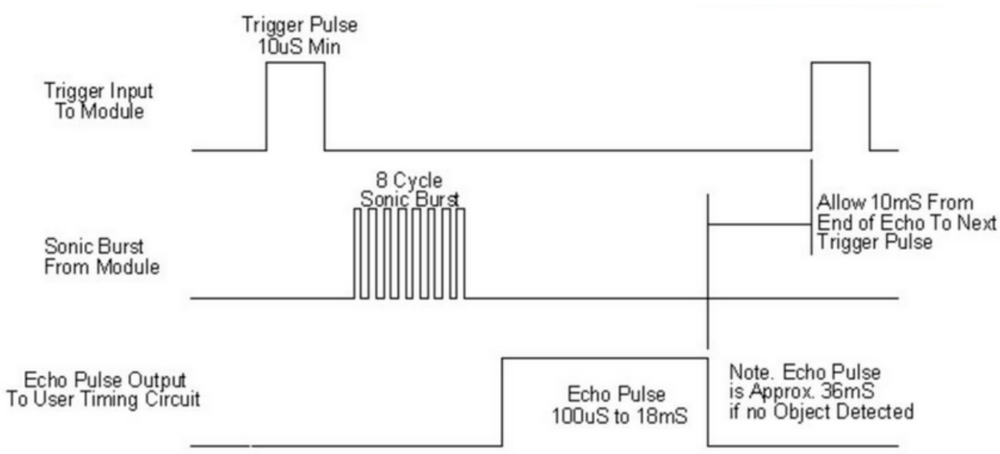


Figure 22: HC-SR04 timing schematic diagram

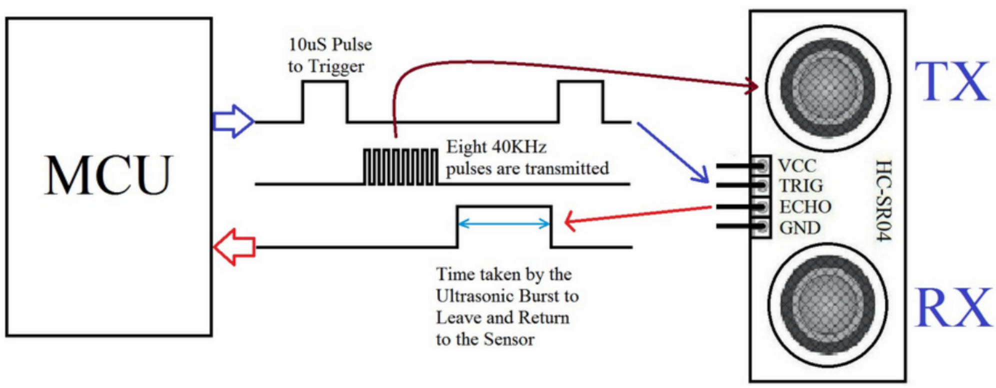


Figure 23: HC-SR04 timing schematic diagram

3.5 IMU sensor: MPU 9050

The MPU-9250 is a compact 9-axis motion tracking sensor developed by TDK InvenSense. It integrates three sensing components into a single chip:`accelerometer, gyroscope, magnetometer`. This module is more superior than older IMU sensor generation like MPU 6050. By integrated 9 axis for 3 sensors in 1 module, it helps simplify electrical system while provide the same functions that requires multiple modules. This combination enables full 9 degrees of freedom (9-DoF) motion tracking, making the MPU-9250 suitable for applications requiring orientation, motion detection, and navigation.

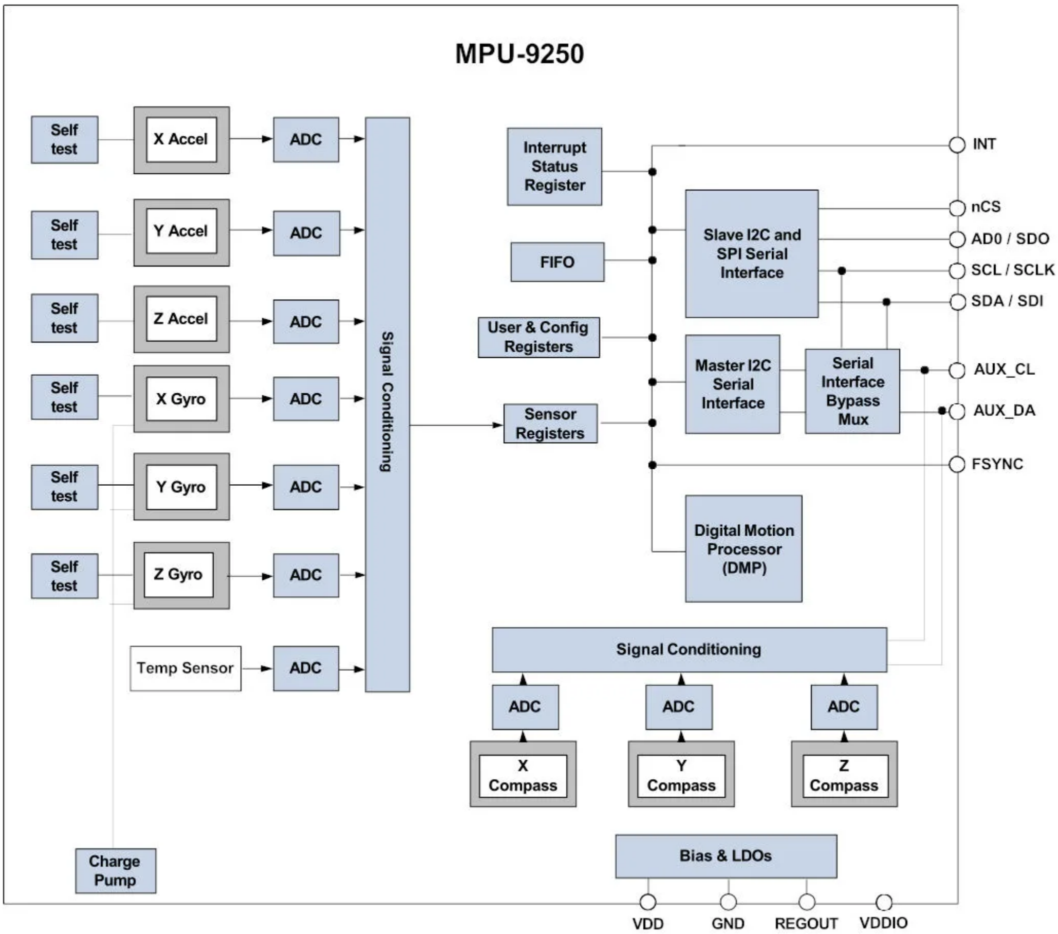


Figure 24: MPU-9250 block diagram, more information can be found in this datasheet, announced by TDK InvenSense

Overview

In this part, we not dive into detailed and technical specification, we only focus 3 main general functions are: **gyroscope, accelerometer and magnetometer**

The MPU-9250 consists of three independent vibratory MEMS rate gyroscopes, which detect rotation about the X-, Y-, and Z- Axes. When the gyros are rotated about any of the sense axes, the Coriolis Effect causes a vibration that is detected by a capacitive pickoff. The resulting signal is amplified, demodulated, and filtered to produce a voltage that is proportional to the angular rate. This voltage is digitized using individual on-chip 16-bit Analog-to-Digital Converters (ADCs) to sample each axis. The full-scale range of the gyro sensors may be digitally programmed to ±250, ±500, ±1000, or ±2000 degrees per second (dps). The ADC sample rate is programmable from 8,000 samples per second, down to 3.9 samples per second, and user-selectable low-pass filters enable a wide range of cut-off frequencies.

The MPU-9250's 3-Axis accelerometer uses separate proof masses for each axis. Acceleration along a particular axis induces displacement on the corresponding proof mass, and capacitive sensors detect the displacement differentially. The MPU-9250's architecture reduces the accelerometers' susceptibility to fabrication variations as well as to thermal drift. When the device is placed on a flat surface, it will measure 0g on the X- and Y-axes and +1g on the Z-axis. The accelerometers' scale factor is calibrated at the factory and is nominally independent of supply voltage. Each sensor has a dedicated sigma-delta ADC for providing digital outputs. The full scale range of the digital output can be adjusted to ±2g, ±4g, ±8g, or ±16g

The 3-axis magnetometer uses highly sensitive Hall sensor technology. The magnetometer portion of the IC incorporates magnetic sensors for detecting terrestrial magnetism in the X-, Y-, and Z- Axes, a sensor driving circuit, a signal amplifier chain, and an arithmetic circuit for processing the signal from each sensor. Each ADC has a 16-bit resolution and a full scale range of ±4800 µT.

4. Monitoring System

For the first version of control and monitoring GUI, we will using QT framework with Python. Main features of the system is to monitor sensors data receive from MAVEN-01, velocity battery level and many other features. This system will information intuitively about the condition of environments surround the rover which is the main target of MAVEN-01.

QT framework is flexible framework for making desktop application, compatible with both Python and C++, more specific, MMS was using Pyside6, a library that allow users use QT APIS through Python.

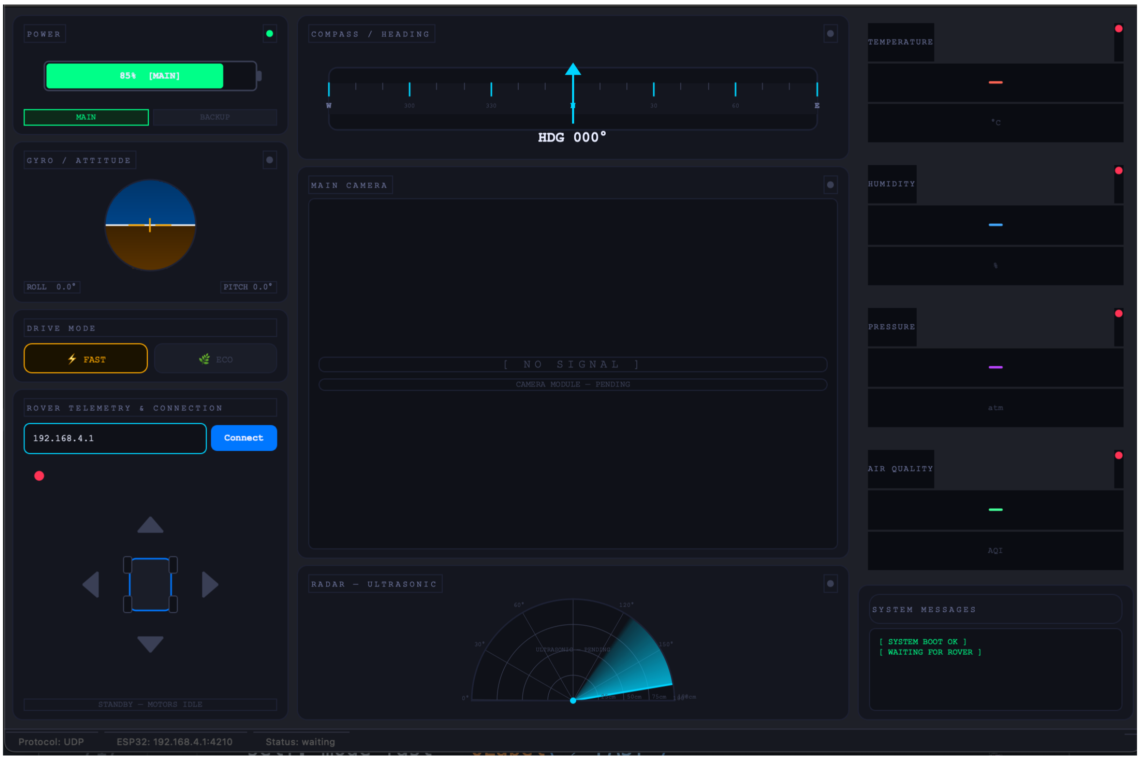


Figure 25: Main user interface of MMS first version

(We will talk about this more later)

4.1 Network communication protocol: MQTT

To connect from MMS to the Rover, we use **MQTT(Messaging Queuing Telemetry Transport)** protocol, a lightweight, publish/subscribe network protocol designed for low-bandwidth, unreliable networks and resource-constrained IoT devices. The key aspects oF MQTT is **publish & subscribe model**: Clients publish data to a "topic" on a broker, and other clients subscribe to that topic to receive the data, decoupling the sender and receiver.MQTT is ideal for remote monitoring, as it requires little bandwidth and has a small code footprint.

MQTT Workflow

Maven-01's main goal is to go to hazardous places and collect environmental information such as pressure, temperature, and humidity. Even though the 2.4 GHz Wi-Fi bandwidth is quite large, it is not very stable when the rover is covered by obstacles. Therefore, we still need MQTT for low-latency and robust data transmission.

In the MQTT workflow diagram 26, all sensor data are sent to a middle distributor block called the **Broker**, which can be related to a simpler example such as Steam.

Steam works like a distribution platform for publishing and selling games. All game studios (Sensors) publish their games (sensor data) to Steam (Broker), and we (Subscribers) go to Steam and buy the game (message) by saying, *"I want this game, and I will pay for it."* (subscribe). Then, Steam sends you the message, *"Ok, I got your money. Here is your game."*, along with a copy of the game (payload).

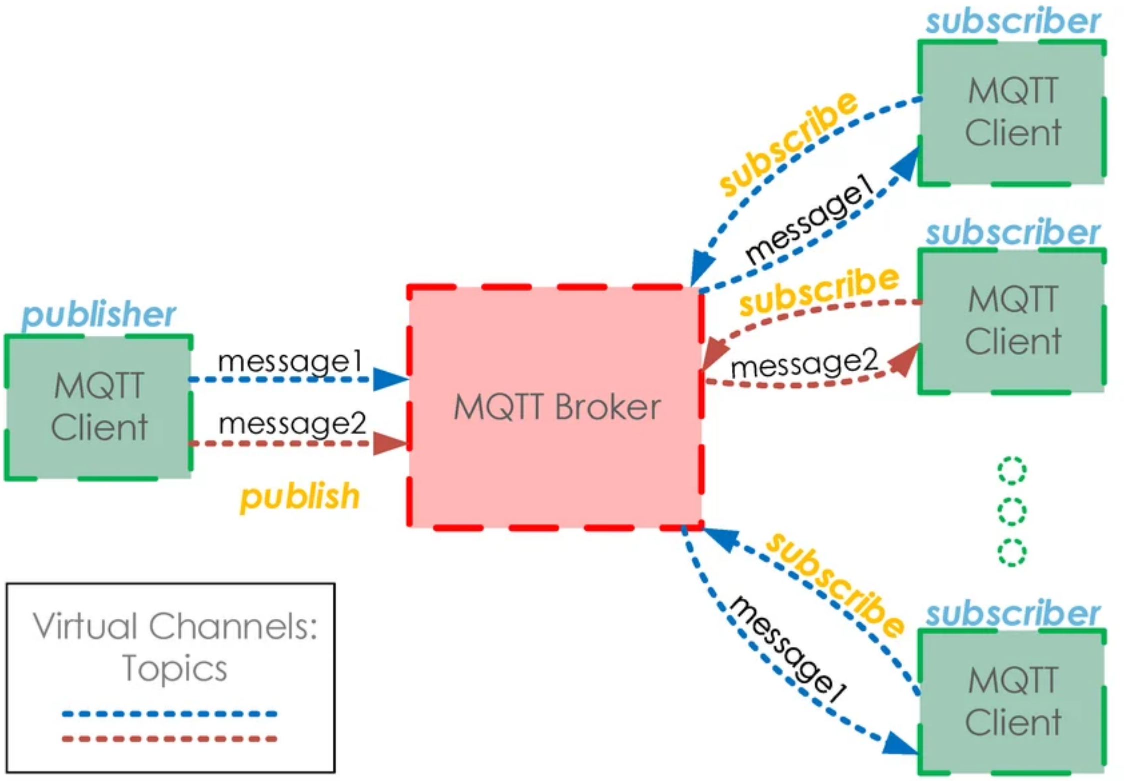


Figure 26: MQTT workflow diagram

5. Conclusions

References

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